

FINAL PROJECT

"Profiling and Characterization of Conversation with a Dual Reinforcement Learning Model"

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DIRECTED BY: GADDI BLUMROSEN



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Holon Institute
of Technology

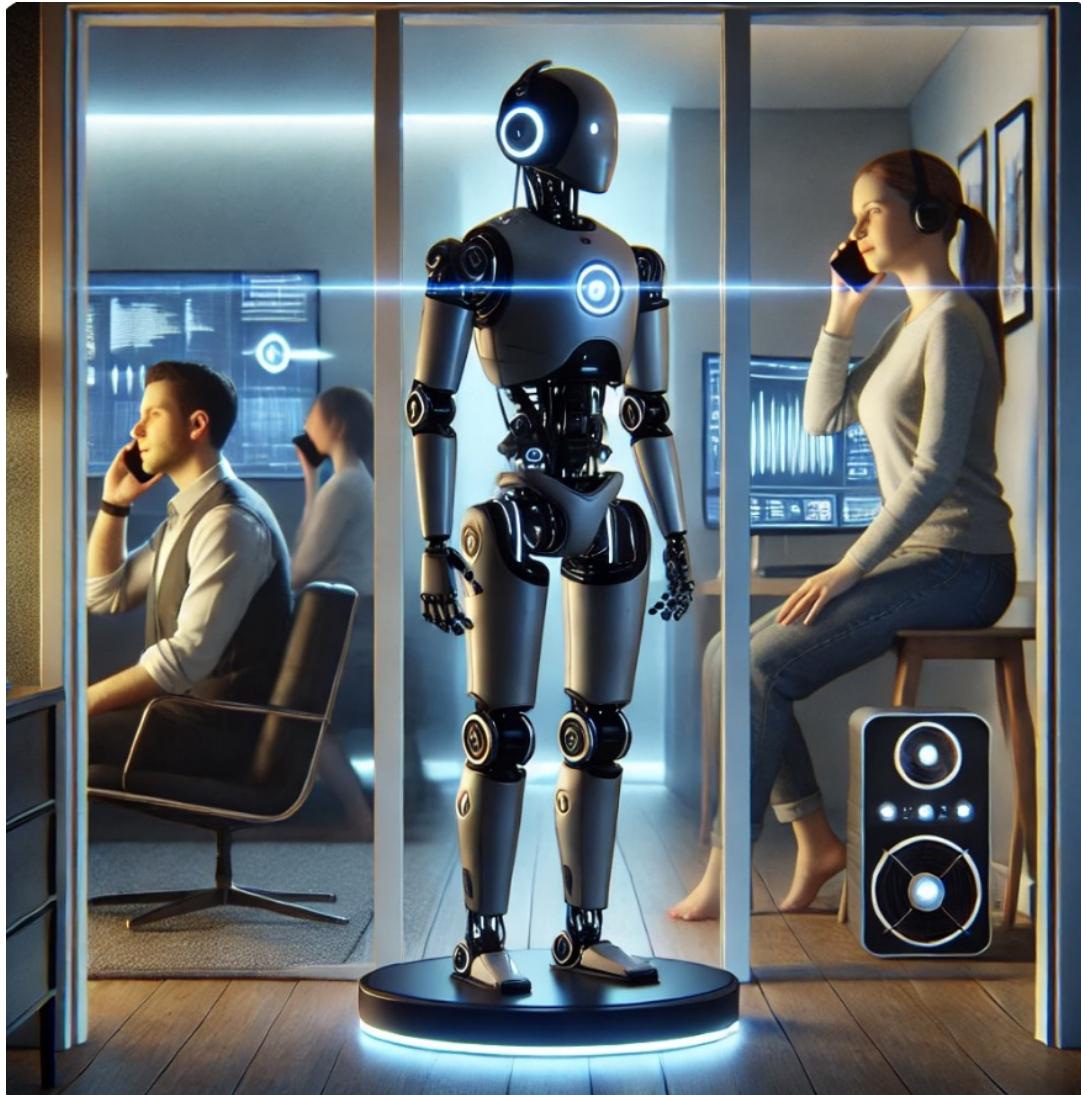


Motivation

Why it is important to qualify conversation quality?

Filed of potentials:

1. Telemedicine
2. Customer service
3. Online education
4. Military Experience Listening



Research Questions

1. How can we effectively characterize conversation quality using multi-modal analysis?
2. Can Multi-Agent Reinforcement Learning (MARL) accurately predict conversation satisfaction?
3. What insights can we gain about factors influencing conversation quality during traumatic experience discussions?

Challenges in Quantifying Interaction Quality

1. Satisfaction Bias
2. Complexity of human communication
3. Dynamic nature of interactions
4. Cultural and contextual variations

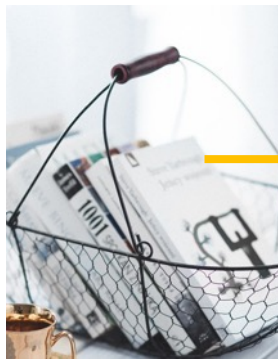


01

Existing Technologies - Traditional Methods



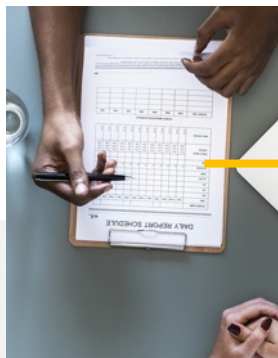
Traditional Methods



Surveys and
Questionnaires



Human
Observation
and Coding



Subjectivity



Lack of
Real-time
Insights:



Surveys and Questionnaires

Types:

1. Likert Scale Surveys: Participants rate statements on a scale (e.g., 1-5 or 1-7), indicating their level of agreement or satisfaction.
2. Open-ended Questionnaires: Allow for more detailed, qualitative responses.
3. Multiple Choice Surveys: Offer predefined options for participants to choose from.

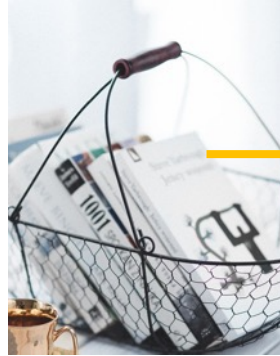
Advantages:

1. Can collect large amounts of data relatively quickly.
2. Easily quantifiable, especially with Likert scales.
3. Can be standardized across different contexts for comparative studies.

Limitations:

1. Subject to response bias (e.g., social desirability bias, acquiescence bias).
2. May not capture nuanced aspects of the interaction.
3. Rely on participants' memory and self-awareness, which can be unreliable.

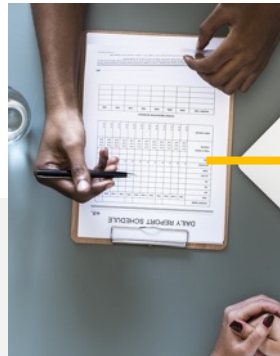
Traditional Methods



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Lack of
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Insights:



Human Observation and Coding



Advantages:

1. Can capture subtle nuances that participants might not self-report.
2. Allows for systematic analysis of interaction patterns.
3. Can provide rich, qualitative data.

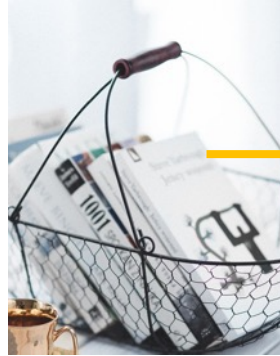
Limitations:

1. Highly time-consuming and labor-intensive.
2. Requires extensive training of observers to ensure reliability.
3. Potential for observer bias or inconsistency between different coders.

Types:

1. Behavioral Coding: Observers note the occurrence of specific behaviors.
2. Interaction Process Analysis (IPA): Categorizes interactions into predefined types.
3. Conversational Analysis: Detailed analysis of conversation structure and dynamics.

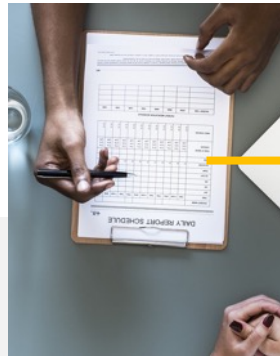
Traditional Methods



Surveys and
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Human
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Subjectivity



Lack of Real-
time Insights:



Subjectivity

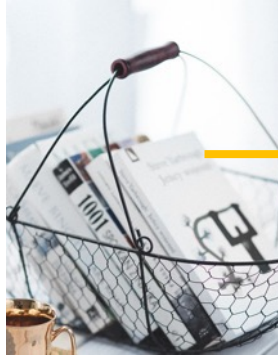


1. Self-report measures are inherently subjective, reflecting personal perceptions rather than objective reality.
2. Different participants may interpret questions or rating scales differently.
3. In observational methods, coders may have subjective interpretations of behaviors or interaction quality.

Hawthorne Effect:

1. Participants may alter their behavior when they know they're being observed or studied.
2. This can lead to artificial improvements in interaction quality that don't reflect natural behavior.
3. The effect can persist even with unobtrusive observation methods, as long as participants are aware of the study.

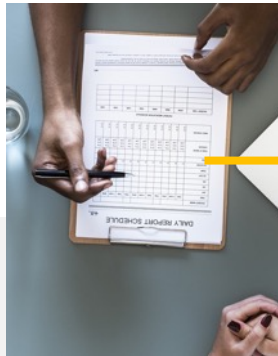
Traditional Methods



Surveys and
Questionnaires



Human
Observation
and Coding



Subjectivity



Lack of Real-
time Insights:



Lack of Real-time Insights:



1. Traditional methods typically assess interactions after they've occurred.
2. This delay limits the ability to provide immediate feedback or interventions.
3. Real-time adjustments to improve interaction quality are not possible with these methods.

02

Modern Techniques



I. Automated audio analysis

II. Text-based analysis:



Automated audio analysis



Automated Text analysis

03

Proposed Approach - Multi-Modality Analysis using MARL



Design criteria & Multi-Modality Analysis

1. Combining audio and text analysis
2. Capturing content and delivery
3. Holistic view of conversation dynamics
4. Exploit multi modal resources
5. Decision in real time



Introduction to MARL

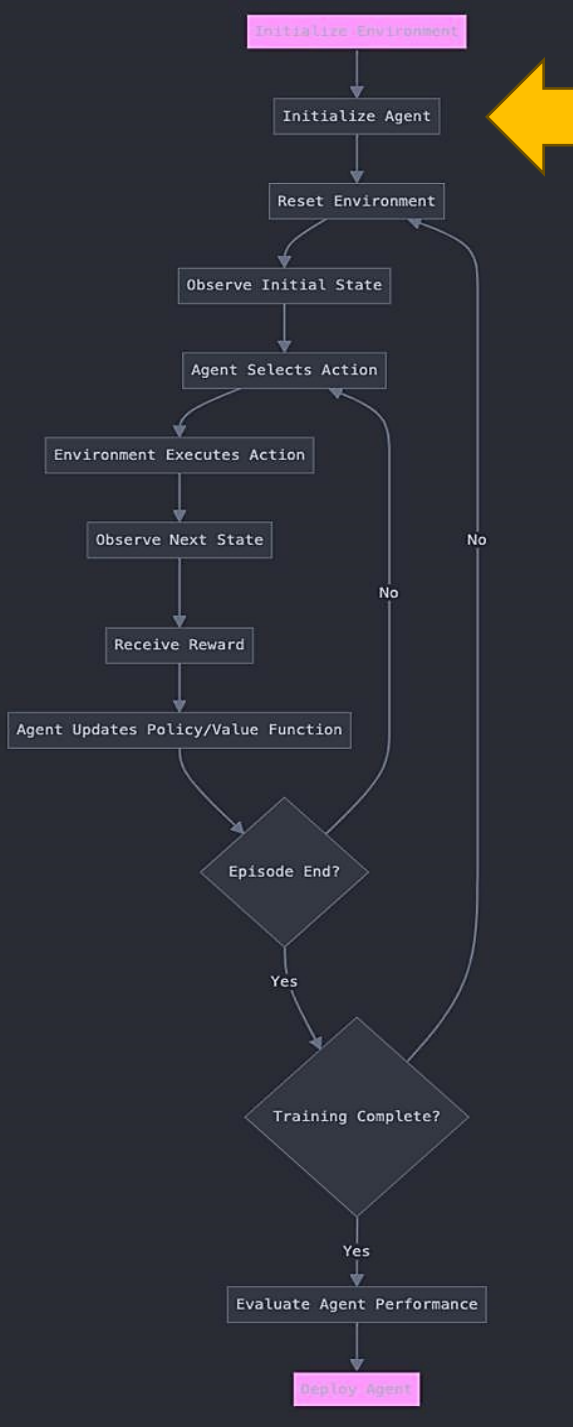
1. Multi-Agent Reinforcement Learning
2. Multiple agents learning in shared environment
3. Modeling dynamic, interactive nature of conversations



Reinforcement Learning

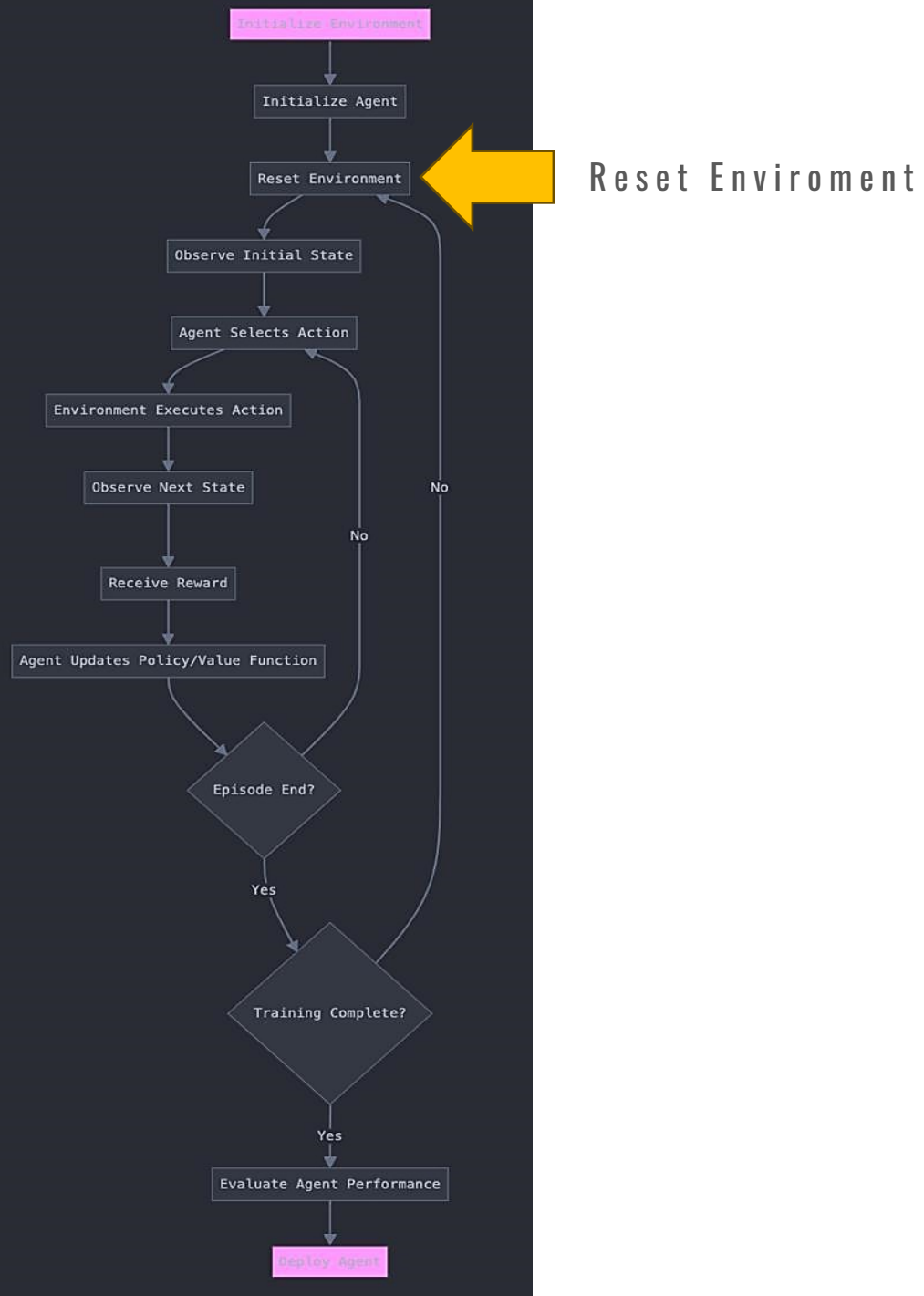
$\theta \leftarrow$ random initialization

Initialize Agent

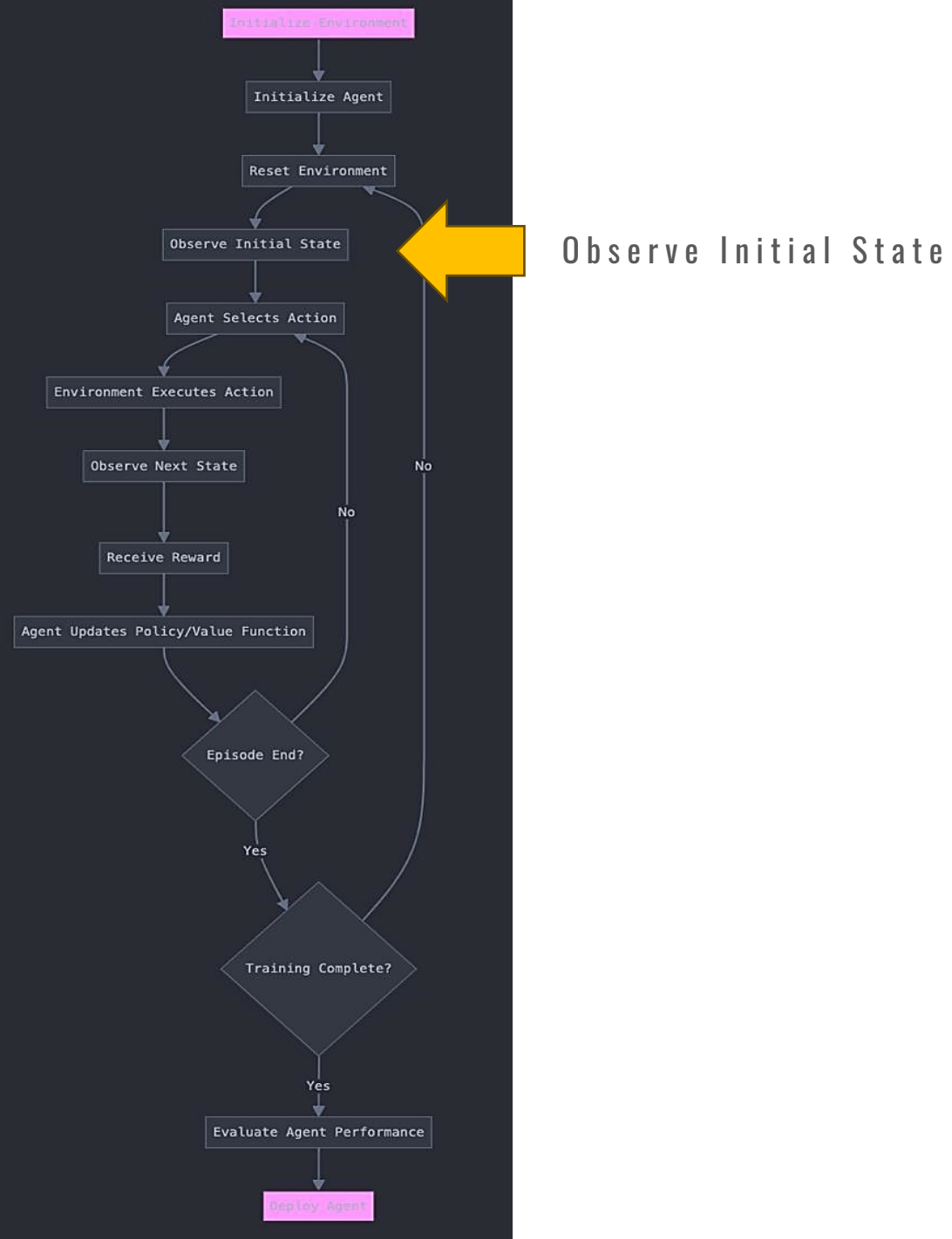


Reinforcement Learning

$$s_0 \sim p(s_0)$$

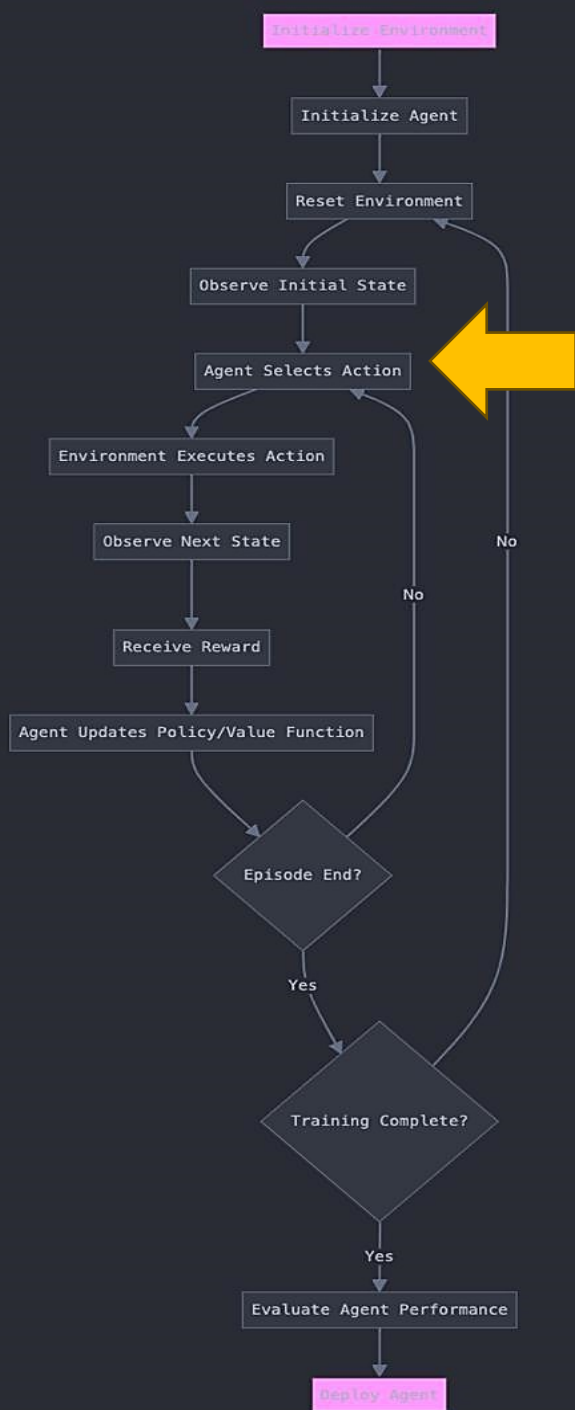


Reinforcement Learning

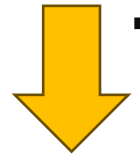


Receive s_0

Reinforcement Learning



Agent Selects Action



Textual Actions

Agent Selects Action

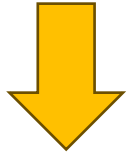
$$a \sim \pi(a|s;\theta)$$

1. Express Empathy
2. Pause for Reflection
3. Ask a Question
4. Share an Anecdote
5. Share an Experience
6. Provide Information
7. Validate Feelings or Opinions
8. Express Disagreement
9. Confirm Understanding
10. Show Agreement
11. Express Gratitude
12. Express Discontent
13. Express Difficulty
14. Close Conversation
15. Give a Compliment
16. Use Humor
17. Express Uncertainty
18. Request Clarification
19. Express Overwhelm
20. Summarize Points
21. Change Topic
22. Offer Assistance or Advice

Textual Actions

Agent Selects Action

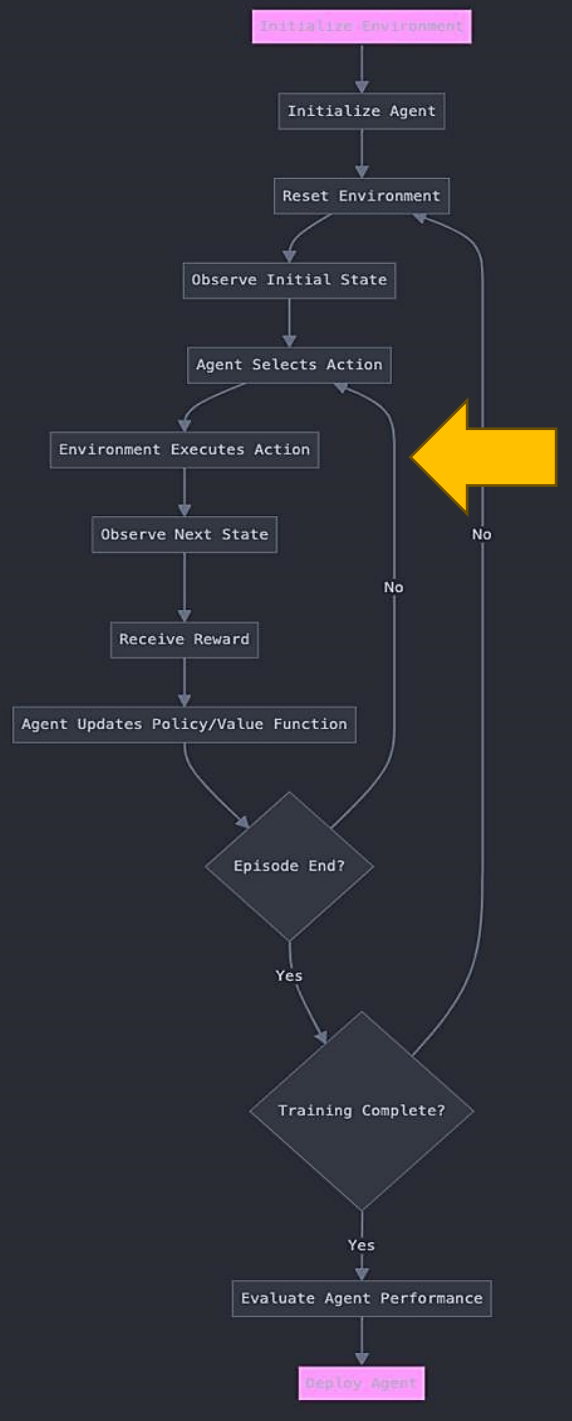
Vocal Actions



1. Express Empathy
2. Pause for Reflection
3. Ask a Question
4. Share an Anecdote
5. Share an Experience
6. Provide Information
7. Validate Feelings or Opinions
8. Express Disagreement
9. Confirm Understanding
10. Show Agreement
11. Express Gratitude
12. Express Discontent
13. Express Difficulty
14. Close Conversation
15. Give a Compliment
16. Use Humor
17. Express Uncertainty
18. Request Clarification
19. Express Overwhelm
20. Summarize Points
21. Change Topic
22. Offer Assistance or Advice

loudness	voice_quality	voice_clarity	voice_variability	speech_rate	voice_texture
Moderate	Percussive	Clear	Moderate	Slow	Moderate
Loud	Tonal	Moderate	Variable	Fast	Plain
Soft	Balanced	Muffled	Consistent	Moderate	Rich

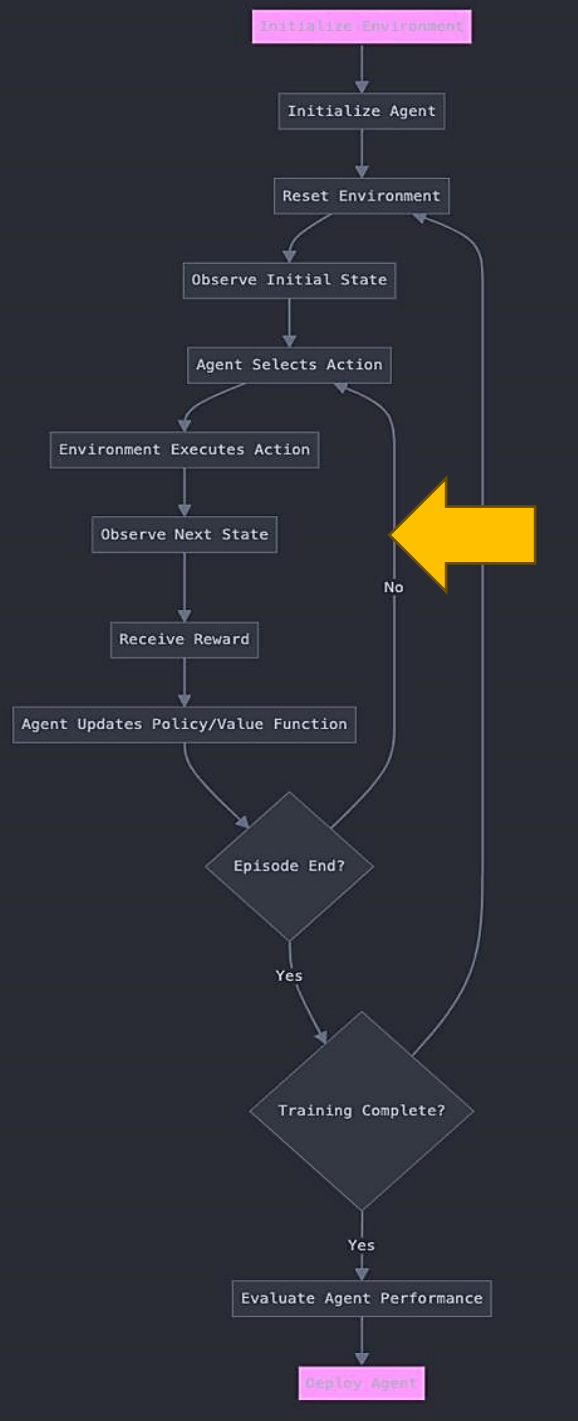
Reinforcement Learning



Environment Executes Action

$$s' \sim P(s'|s,a)$$

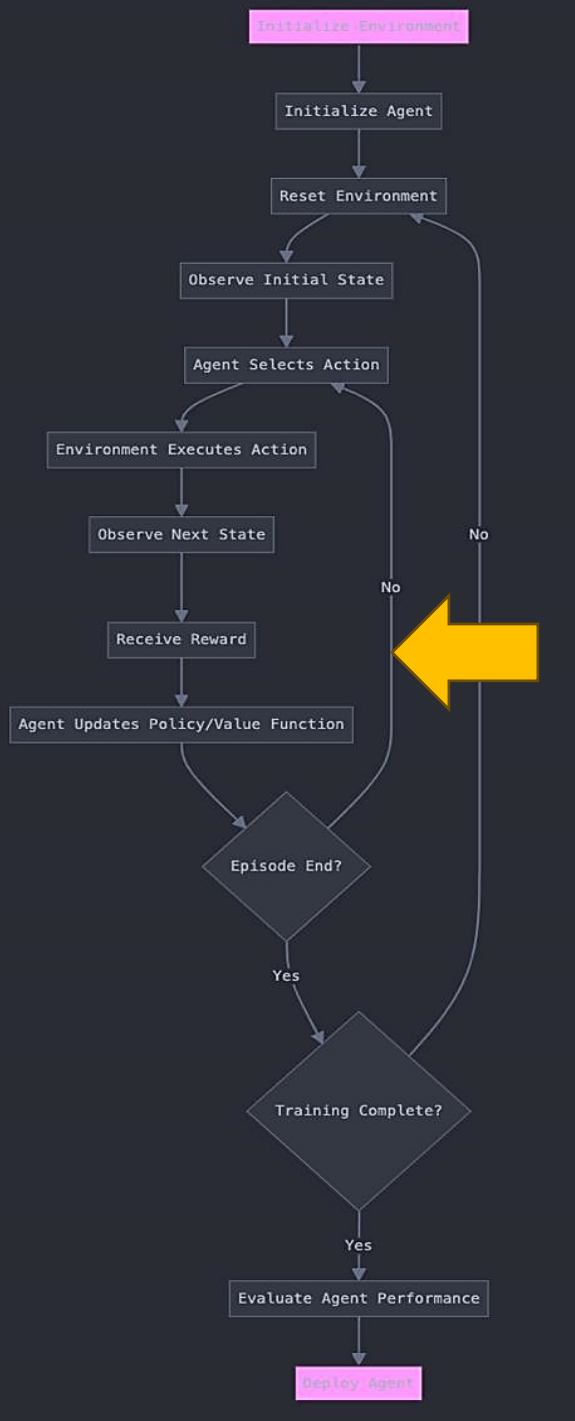
Reinforcement Learning



Observe Next State

Receive s'

Reinforcement Learning



Recive Reward

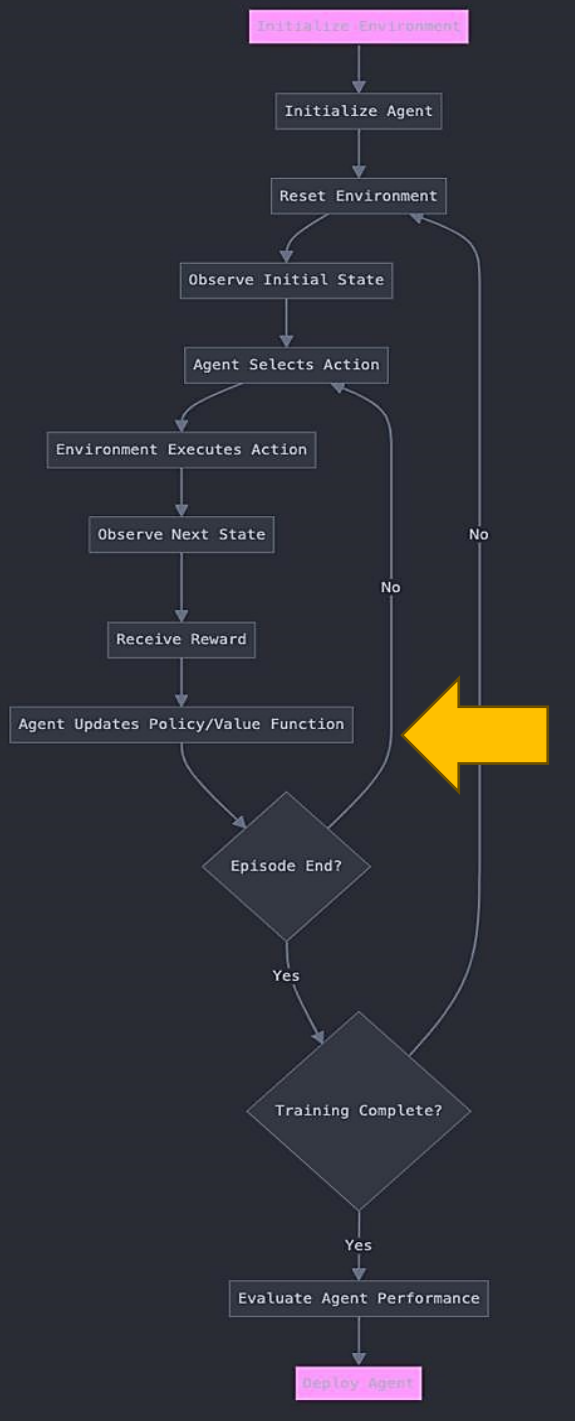
$$r = R(s,a,s')$$

Reinforcement Learning

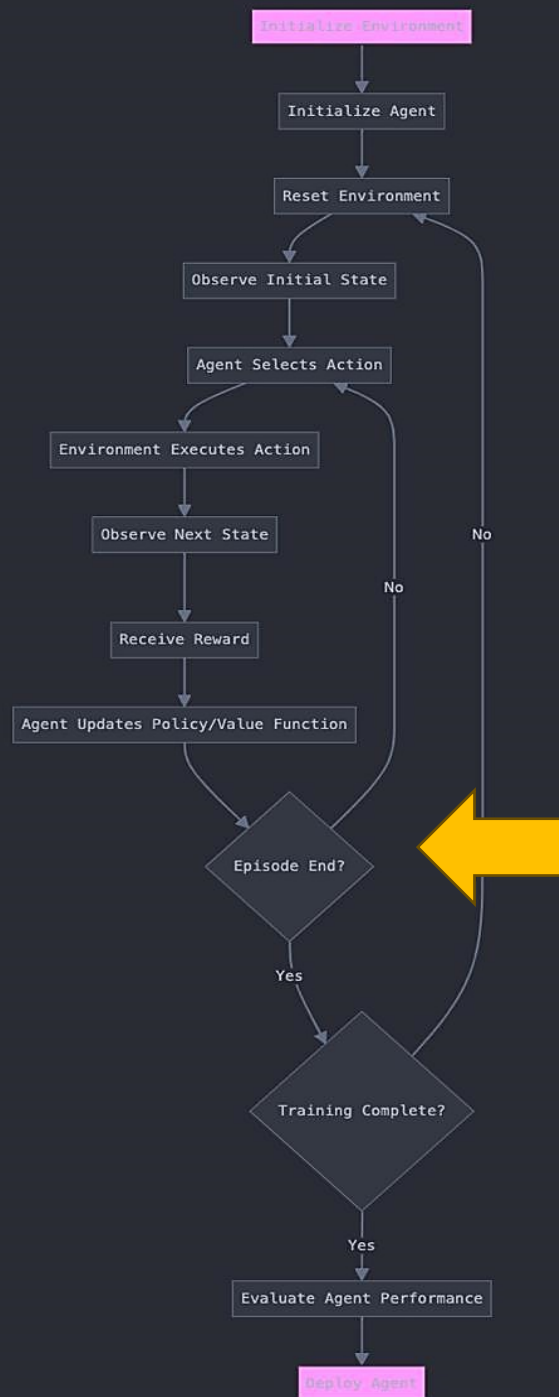
Policy Gradient: $\theta \leftarrow \theta + \alpha \nabla \theta J(\theta)$

Q-learning: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

Agent Updates Policy/Value Function

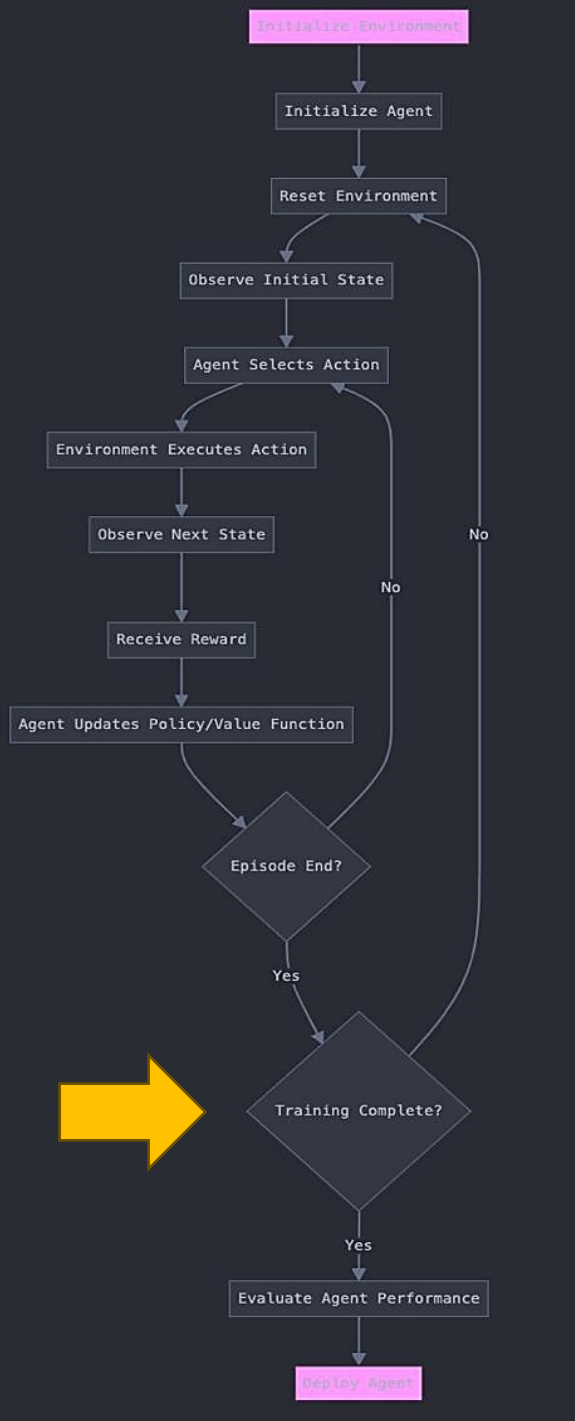


Reinforcement Learning



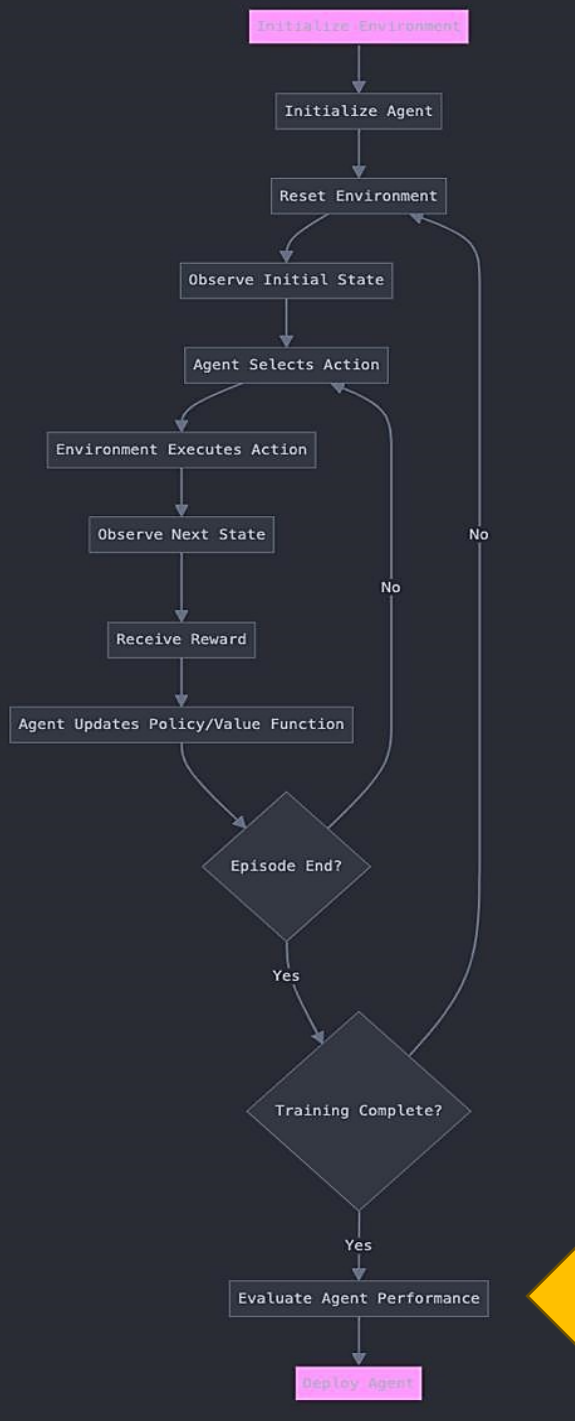
Episode End

Reinforcement Learning



Training Complete

Reinforcement Learning



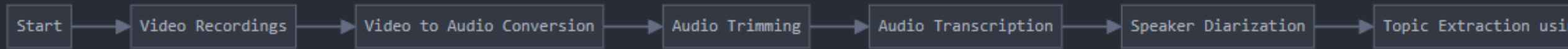
Evaluate Agent Performance

$$J(\pi) = E[\sum \gamma^t r_t]$$

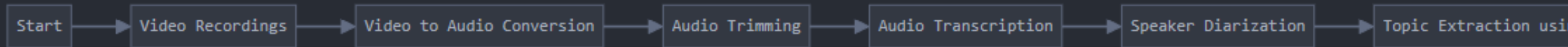
04 Pre Processing



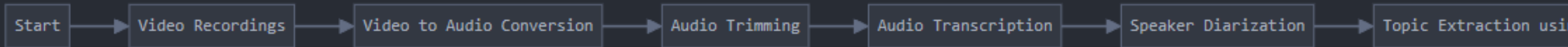
Video Recording – Ariel University



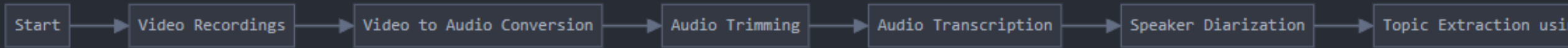
Transform The Video To Mp3



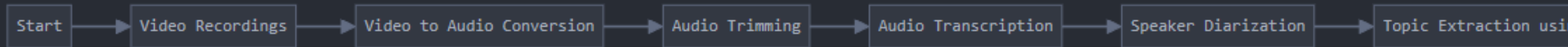
Trimming the video to get the right segments



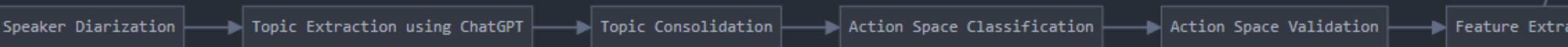
Transcribe Using AWS



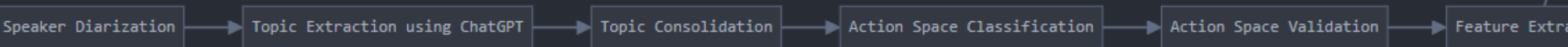
Diarization Using AWS



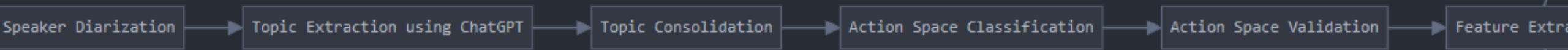
Create Actions Space



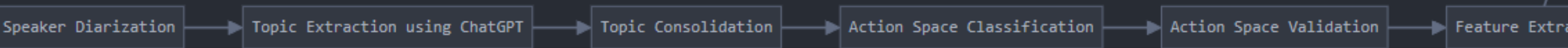
Sharp The Actions Space

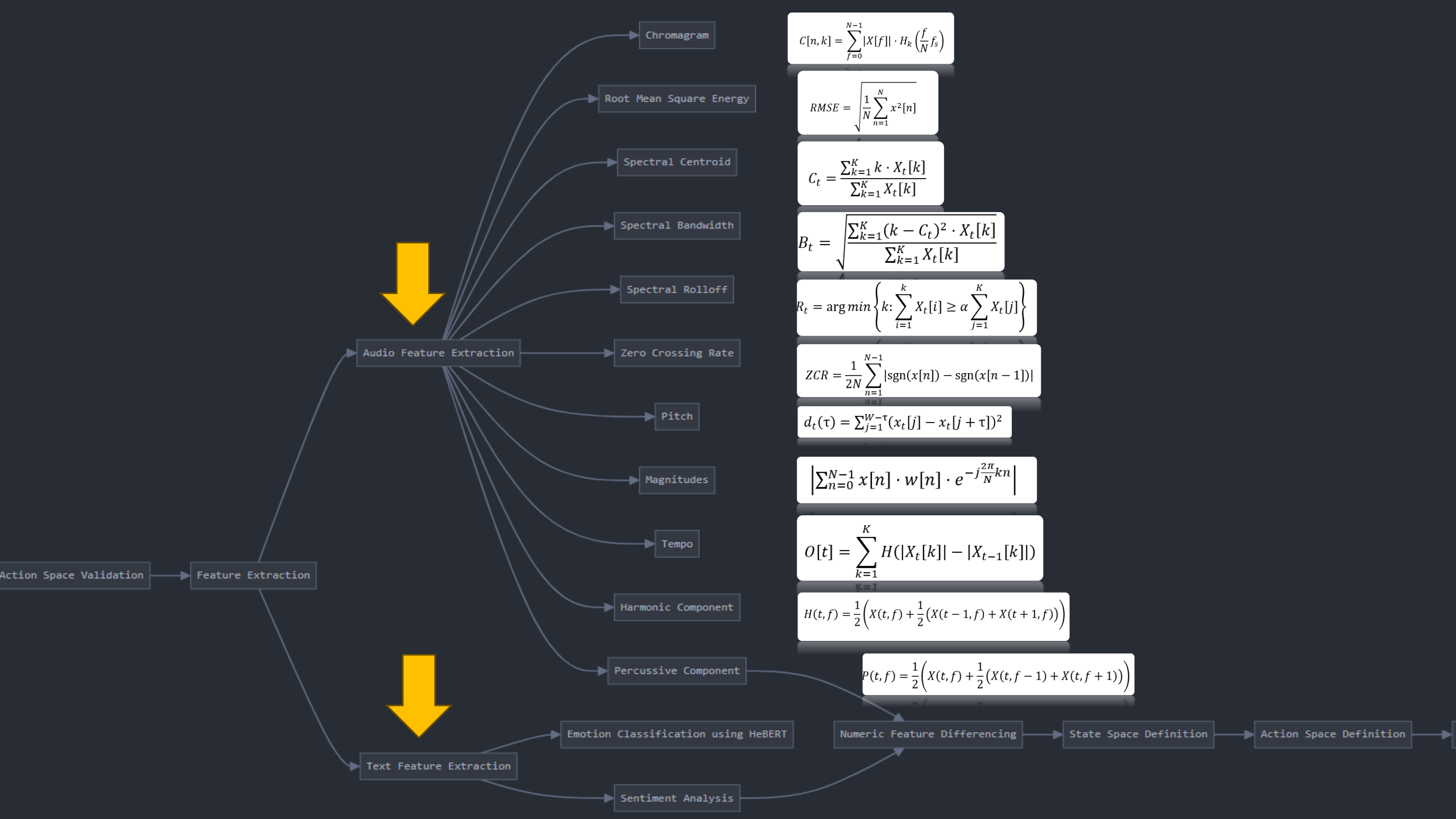


Classified into 22 distinct action

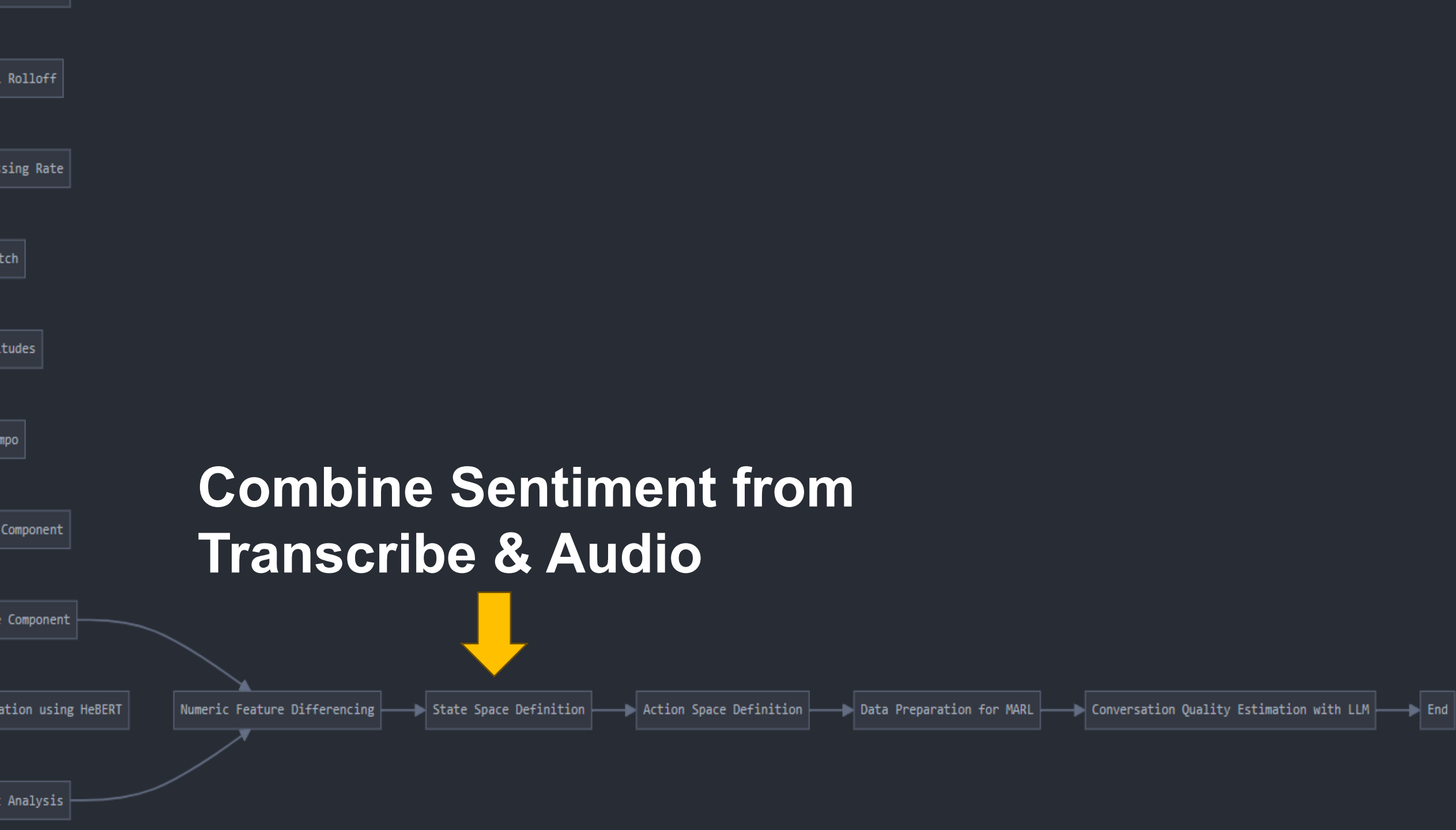


Human Validate

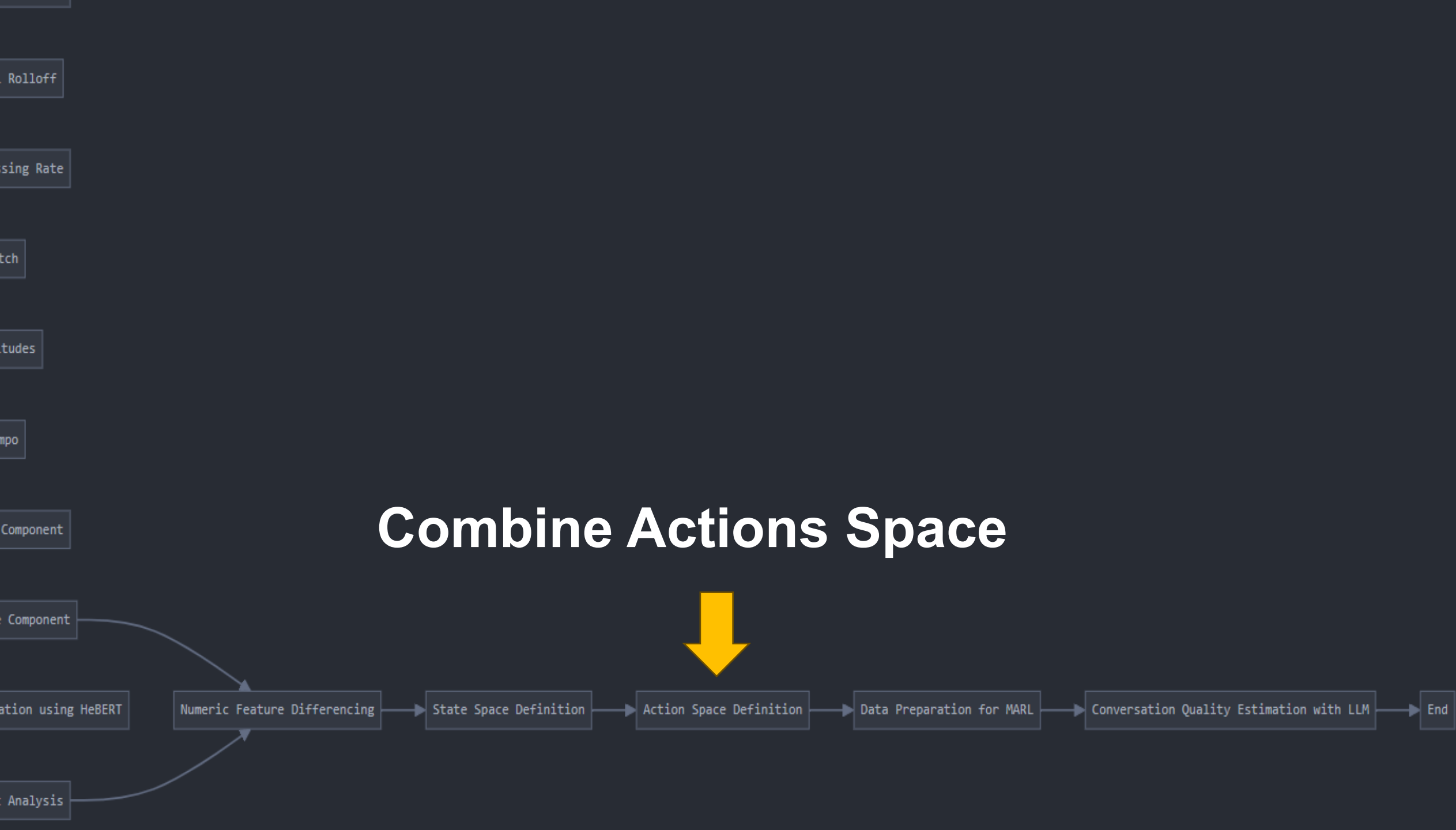




Combine Sentiment from Transcribe & Audio



Combine Actions Space



Rolloff

ing Rate

tch

tudes

mpo

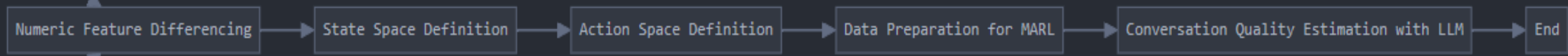
Component

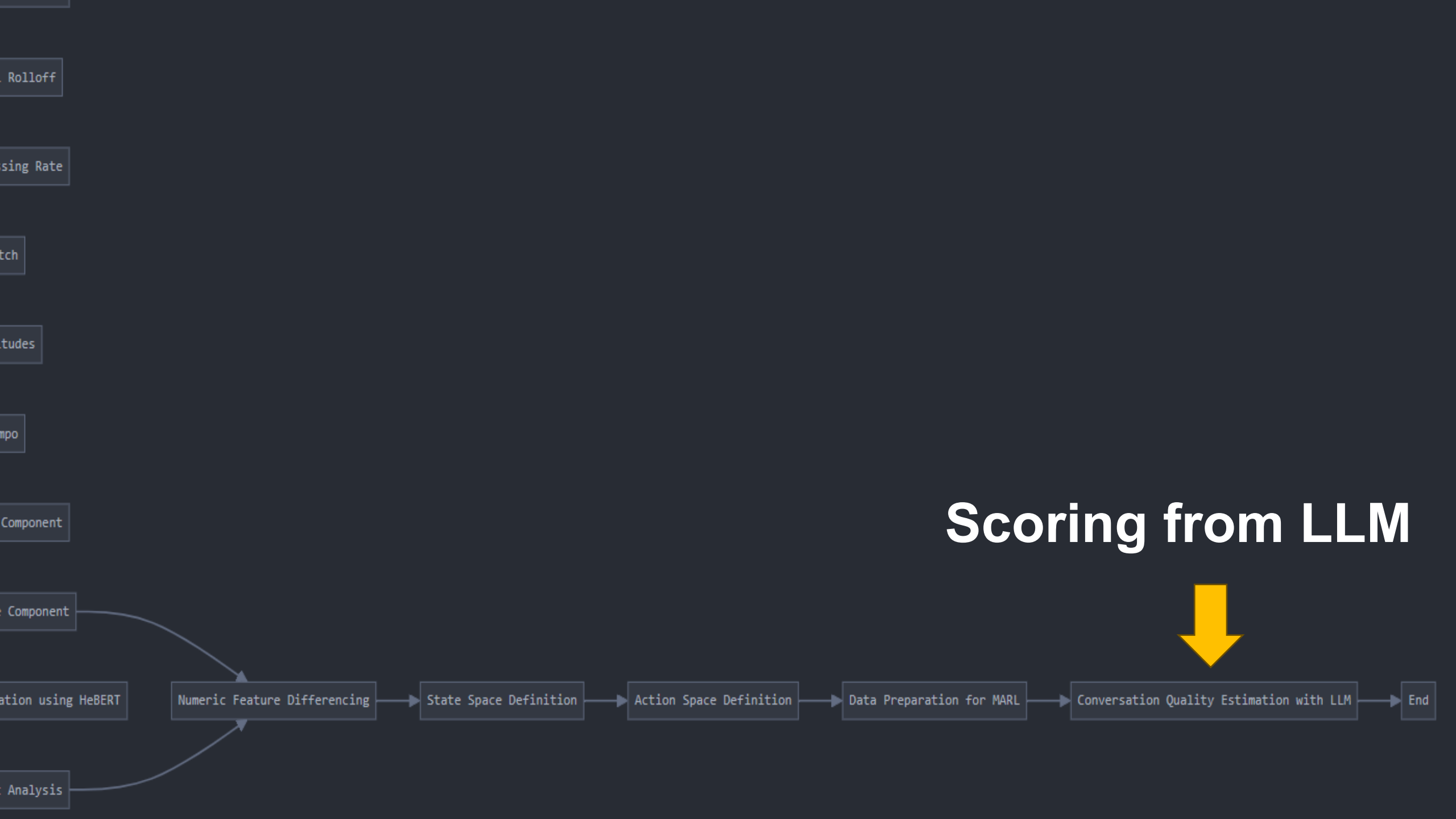
Component

ation using HeBERT

Analysis

Formatting The Data





The MARL Models Used

01

IQL

02

MADDPG

03

MAPPO

04

LLM

01

Independent Q-Learning (IQL)

IQL

1. **Separate Q-function** for each agent
2. **Decentralized learning approach**
3. **State space:** Audio and text features of current speaker
4. **Action space:** Conversational moves
5. **Advantages:** Simplicity, scalability
6. **Limitations:** Lack of explicit coordination

01

IQL

Independent Q-Learning (IQL)

1

Q-Learning Function

In IQL, each agent computes its reward independently using the classical Q-learning function:

$$Q(s_t, a_t) = (1 - \alpha)Q(s_t, a_t) + \alpha \left(r_t + \gamma \max_a Q(s_{t+1}, a) \right)$$

Where:

- r_t is the immediate reward the agent receives for action a_t ,
- α is the learning rate,
- γ is the discount factor,
- s_{t+1} is the future state.

Policy Update

In IQL, the policy follows a greedy policy based on maximizing the Q-function for each state:

2

$$\pi(a|s) = \arg \max_a Q(s, a)$$

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Multi-Agent Deep Deterministic Policy Gradient (MADDPG)

MADDPG

1. **Centralized training, decentralized execution**
2. **Actor-Critic architecture**
3. **State space:** Global state including both speakers
4. **Action space:** Continuous representation of conversational moves
5. **Advantages:** Handles continuous action spaces, considers other agents
6. **Limitations:** Increased complexity, potential instability in training

Multi-Agent Deep Deterministic Policy Gradient (MADDPG)

02

MADDPG

Policy Gradient Update

The policy parameters are updated using the gradient of the cumulative reward:

$$\nabla_{\theta_i} J(\mu_{\theta_i}) = \mathbb{E}_{s_t} [\nabla_{\theta_i} Q_i(s_t, a_1, \dots, a_N)]$$

3

1

Policy

In MADDPG, each agent i operates with a continuous policy $\mu_{\theta_i}(s_t)$, which is a deterministic policy:

$$a_i = \mu_{\theta_i}(s_t)$$

Where:

- s_t is the shared state of all agents,
- a_i is the action of agent i .

Q-Function

The core function for calculating the reward for each agent considers the actions of all other agents:

$$Q_i(s_t, a_1, \dots, a_N) = \mathbb{E} [r_i + \gamma Q_i(s_{t+1}, a'_1, \dots, a'_N)]$$

2

Where:

- r_i is the reward of agent i ,
- $a_1, \dots, a_N$ are the actions of all agents.

Multi-Agent Deep Deterministic Policy Gradient (MADDPG)

02

MADDPG

1. DDPG :

$$L(\theta^Q) = \mathbb{E}_{s_t, a_t, r_t, s_{t+1}} \left[(Q_{\theta^Q}(s_t, a_t) - y_t)^2 \right]$$

$$\nabla_{\theta^\pi} J \approx \mathbb{E}_{s_t} \left[\nabla_a Q_{\theta^Q}(s_t, a) \Big|_{a=\pi_{\theta^\pi}(s_t)} \nabla_{\theta^\pi} \pi_{\theta^\pi}(s_t) \right]$$

2. MADDPG:

$$L(\theta_i^Q) = \mathbb{E} \left[\left(Q_{\theta_i^Q}(o_t^i, a_t^1, \dots, a_t^N) - y_t^i \right)^2 \right]$$

$$\nabla_{\theta_i^\pi} J \approx \mathbb{E} \left[\nabla_{a_i} Q_{\theta_i^Q}(o_t^i, a_t^1, \dots, a_t^N) \nabla_{\theta_i^\pi} \pi_{\theta_i^\pi}(o_t^i) \right]$$

The MARL Models Used

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Multi-Agent Proximal Policy Optimization (MAPPO)

03

MAPPO

1. **Extension of PPO to multi-agent settings**
2. **Trust region policy optimization**
3. **State space:** Local observations + global state
4. **Action space:** Discrete conversational moves
5. **Advantages:** Sample efficiency, stability, easier hyperparameter tuning
6. **Limitations:** May converge to suboptimal policies in some scenarios

Multi-Agent Proximal Policy Optimization (MAPPO)

03

MAPPO

3

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r(\theta) \hat{A}_t, \text{clip} \left(r(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right]$$

Where $\hat{A}_t$ is the advantage estimate for the action at time t , and ϵ defines the clipping range.

Value Function

The value function used to compute the reward is:

$$V^\pi(s_t) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \right]$$

Where r_t is the immediate reward and γ is the discount factor.

1

Policy

In MAPPO, each agent optimizes its policy by calculating the ratio between the current policy and the old policy.

The policy is denoted as π_θ , where θ are the parameters of the policy:

$$\pi_\theta(a_t|s_t) = P(a_t|s_t;\theta)$$

Where:

- a_t is the action at time t ,
- s_t is the current state.

Loss Function

The ratio between the new policy and the old one is denoted as $r(\theta)$ and is defined as:

2

$$r(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

Multi-Agent Proximal Policy Optimization (MAPPO)

03

MAPPO

1. PPO :

$$L^{PPO}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

$$V(s_t) - \gamma V(s_{t+1}) + {}_t r = \hat{A}_t$$

2. MAPPO:

$$L^{MAPPO}(\theta_i) = \mathbb{E}_t \left[\min \left(r_t^i(\theta_i) \hat{A}_t^i, \text{clip}(r_t^i(\theta_i), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^i \right) \right]$$

$$V(s_t^i, s_t^j) - \gamma V(s_{t+1}^i, s_{t+1}^j) + {}_t^j w_j r \sum_{j \neq i} + {}_t^i r = \hat{A}_t^i$$

The MARL Models Used

01

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MAPPO

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LLM

What are the differences?

01 IQL

02 MADDPG

03 MAPPO

Aspect	IQL	MADDPG	MAPPO
Full Name	Independent Q-Learning	Multi-Agent Deep Deterministic Policy Gradient	Multi-Agent Proximal Policy Optimization
Base Algorithm	Q-Learning	Extends DDPG (Deep Deterministic Policy Gradient)	Extends PPO (Proximal Policy Optimization)
Policy Type	Value-based (implicit policy)	Deterministic	Stochastic
Learning Approach	Off-policy	Off-policy	On-policy
Network Architecture	Single Q-network per agent	Actor-Critic (separate networks)	Policy and Value networks (can be shared)
Centralized Component	None (fully decentralized)	Critic network	Value function
Action Space	Typically discrete, can be adapted for continuous	Primarily continuous, can be adapted for discrete	Both continuous and discrete
Stability	Can be unstable due to overestimation bias	Can be less stable, sensitive to hyperparameters	Generally more stable due to trust region constraint
Sample Efficiency	Often most sample efficient	Can be more sample efficient due to experience replay	May require more samples, but uses them more effectively
Scalability	Scales well with number of agents	Can struggle with large numbers of agents due to critic complexity	Generally scales better to more agents than MADDPG

04

LLM

The MARL Models Used

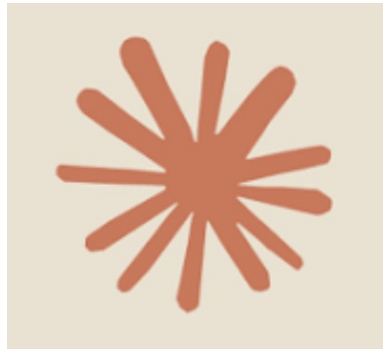
Micro Conversation

04

LLM

The MARL Models Used

Micro Conversation



Micro Conversation

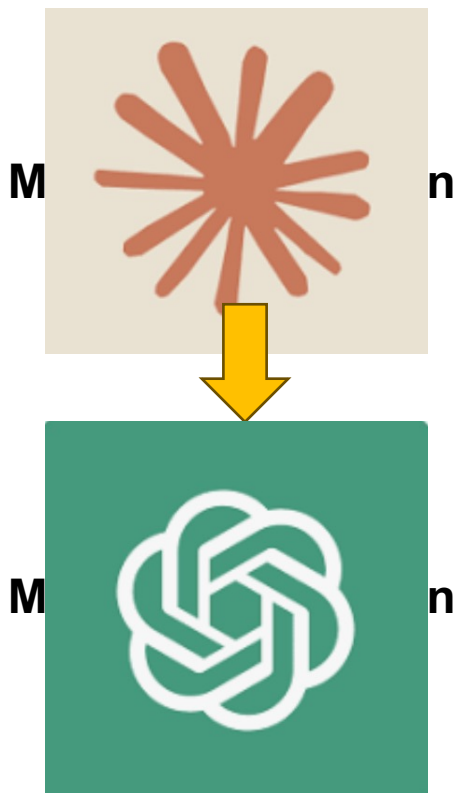


04

LLM

The MARL Models Used

Prompt – Classification for 22 Textual Actions



04

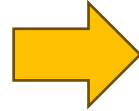
LLM

The MARL Models Used

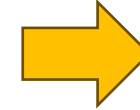
Prompt – Classification for 22 Textual Actions



Classification



Scoring

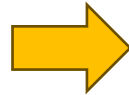


Classes:

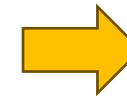
Pleased or Not



Classification



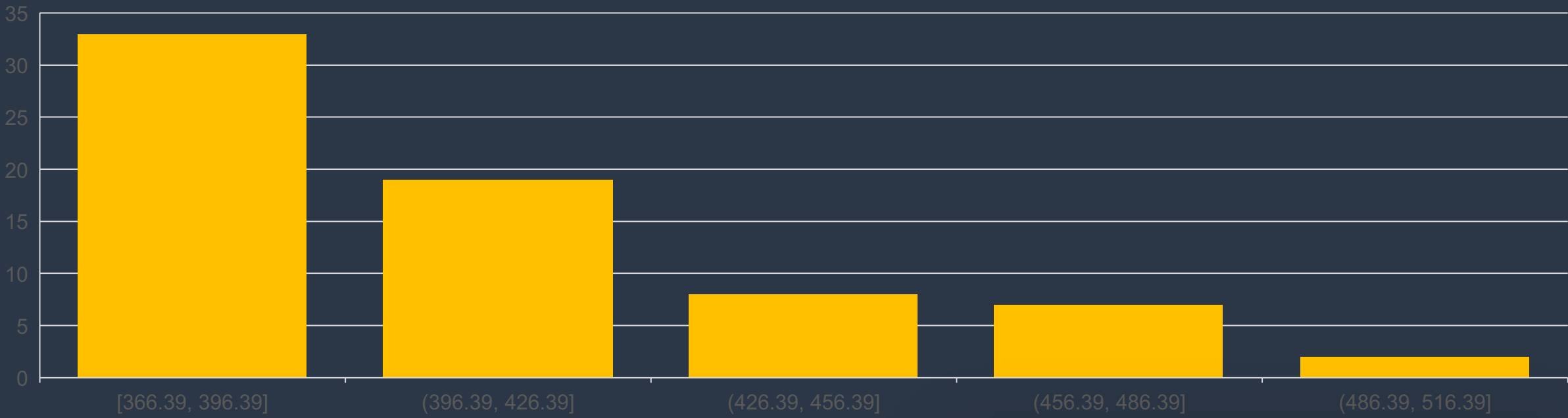
Scoring



Pleased

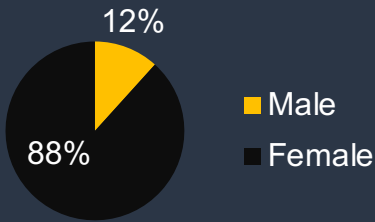
Dataset Overview

Conversation Time Duration



Ages Range	Number of Dyads	Peticipents	Action Space
19-30 years	69 video-recorded conversations	138	22 Actions
Conversation Duration	Conversation Parts	Subject Of a Conversation	Video Trimmed Time
~25 minutes	2	COVID-19 quarantine experiences and return to normalcy	~5-7 minutes

Gender



Dataset

Arial University -
Students

05

RESULTS



- I. Model Performance Comparison
- II. Cumulative Rewards and Learning Curves
- III. Feature Importance Analysis
- IV. Model Comparison with LLMs

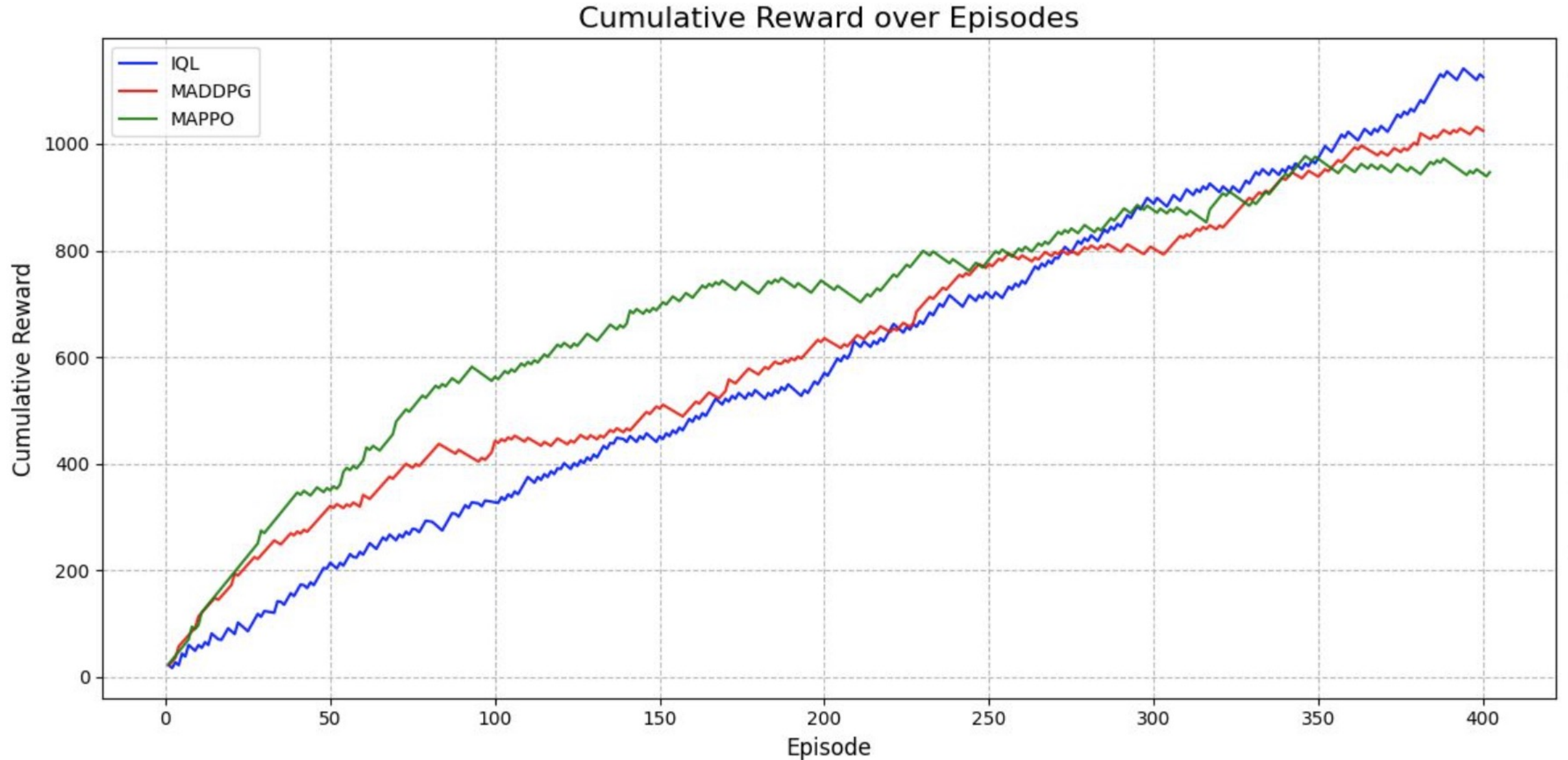
1. Model Performance Comparison

RL Method	TEST RMSE	VALIDATION RMSE	MAPE	Correlation	P-Value
MADDPG	2.90	2.56	6.92	0.64	0.049
MAPPO	7.58	8.24	22.9	-0.62	0.041
IQL	1.66	0.78	4.49	0.71	0.021

LLM Method	MAPE
ChatGPT (Israeli)	28.9
ChatGPT (American)	27.8
Calude (Israeli)	21.2
Claude (American)	32.07

2. Cumulative Rewards and Learning Curves

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2. Measurement – Pleased or not Pleased

3. Classification Reports

$$x_{Min-Max\ Feature\ Scaling} = 2 \times \frac{x - \min(score)}{\max(score) - \min(score)} - 1$$

"pleased" if $x_{MinMax} \geq 0$

"not pleased" if $x_{MinMax} < 0$

IQL Classification Report:

	precision	recall	f1-score	support
not pleased	0.67	1.00	0.80	6
pleased	1.00	0.62	0.77	8
accuracy			0.79	14
macro avg	0.83	0.81	0.78	14
weighted avg	0.86	0.79	0.78	14

Figure 17 - classification report for IQL model

MAPPO Classification Report:

	precision	recall	f1-score	support
not pleased	0.40	0.33	0.36	6
pleased	0.56	0.62	0.59	8
accuracy			0.50	14
macro avg	0.48	0.48	0.48	14
weighted avg	0.49	0.50	0.49	14

Figure 18 - classification report for MAPPO model

MADDPG Classification Report:

	precision	recall	f1-score	support
not pleased	0.50	0.33	0.40	6
pleased	0.60	0.75	0.67	8
accuracy			0.57	14
macro avg	0.55	0.54	0.53	14
weighted avg	0.56	0.57	0.55	14

3. Classification Reports

IQL Classification Report:

	precision	recall	f1-score	support
not pleased	0.67	1.00	0.80	6
pleased	1.00	0.62	0.77	8
accuracy			0.79	14
macro avg	0.83	0.81	0.78	14
weighted avg	0.86	0.79	0.78	14

Figure 17 - classification report for IQL model

MAPPO Classification Report:

06 Conclusion



Conclusions

1. **MARL models outperformed LLMs in conversation quality assessment**
 1. IQL: MAPE 4.49%
 2. Best LLM (Claude): 0.16 Pearson correlation
2. **IQL showed strongest performance among MARL models**
3. **MADDPG and MAPPO demonstrated promise in modeling complex interactions**
 1. MAPPO exhibited most stable learning curve
4. **Multi-modal analysis (audio + text) proved more effective than text-only approaches**
5. **Cultural context influences conversation assessment Slight variations observed between Israeli and American perspectives**

Limitations

1. Limitations and Future Work:

1. Expand dataset beyond 69 conversations
2. Explore wider range of topics beyond research participants discussions
3. Investigate performance across diverse cultural contexts
4. Develop models for long-term conversational dynamics
5. Incorporate additional modalities (e.g., visual cues)